MADEIHA ARSHAD

Software Developer | Backend & Full-Stack

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PROFESSIONAL SUMMARY

Final-year Software Engineering student at Bahria University Islamabad (graduating June 2027) with backend internship experience in NestJS, TypeORM, and PostgreSQL, and hands-on expertise in full-stack web development (MERN), AI/computer vision (YOLOv8, MediaPipe), and cloud deployment. Built and deployed production-grade applications, including SafeCity AI. Seeking a remote or part-time software developer role where I can contribute to real-world products while completing my degree.

PROFESSIONAL EXPERIENCE

Backend Developer Intern | Sybrid Pvt. Ltd., Islamabad | Jul 2026 – Aug 2026 (2 months)

- Developed backend modules for a Hospital Management System using NestJS, TypeORM, and PostgreSQL.
- Built a real-time chat system using NestJS WebSocket gateways with PostgreSQL message persistence.
- Implemented transactional database operations using TypeORM QueryRunner to keep data consistent across multi-step operations.
- Implemented JWT-based authentication and role-based access control to secure sensitive medical data and user roles.
- Designed and optimized database schemas.
- Tested APIs with Postman.

EDUCATION

Bachelor of Science – Software Engineering 2023 – Present (Expected June 2027)

Bahria University, Islamabad

- CGPA: 3.29 / 4.0
- Relevant Coursework: OOP, DSA, DBMS, OS, Software Design & Architecture, Web Engineering, Cloud Computing, Software Quality Engineering (SQE), HCI, AI, Technical Writing

FSC Pre-Engineering | Bahria College Zafar Campus, E-8 | 2022

TECHNICAL SKILLS

Languages: Python, JavaScript, Java, C++, C#

Frontend: React.js, Next.js, HTML5, CSS3, Tailwind CSS, Framer Motion, Vite, Responsive Design

Backend: Node.js, NestJS, Express.js, Spring Boot, Flask, TypeORM, REST APIs, JWT Authentication, Socket.io, WebSockets

AI / ML / CV: YOLOv8, MediaPipe, OpenCV, TensorFlow, Hugging Face APIs, KNN, Decision Tree, Linear Regression

Databases: MongoDB Atlas, MySQL, PostgreSQL, Supabase, Oracle DB, SSMS

Cloud & DevOps: Docker, Railway, Vercel, LocalStack, Terraform (basics), GitHub Actions, CI/CD pipelines

SQA & Testing: Selenium WebDriver, JUnit, Jira AIO, Test Case Design, Bug Reporting

UI/UX: Lo-fi & Hi-fi Prototyping, Wireframing.

Tools: VS Code, Git/GitHub, Eclipse, IntelliJ IDEA, Scene Builder, Postman, draw.io, WPS Office, Figma .

Concepts: OOP, Design Patterns, UML, MVC Architecture, Agile, UWE Web Modeling.

PROJECTS

SafeCity AI – Intelligent Urban Traffic Safety System | React, Node.js, Express, MongoDB Atlas, Python, Flask, YOLOv8, Socket.io, Docker, Vercel, Railway

- Built real-time traffic violation detection platform with AI object detection (YOLOv8), risk scoring, and Dijkstra safe-route planning algorithm.
- Designed and wireframed UI/UX (Lofi and Hifi prototypes)
- Designed and implemented live WebSocket dashboard for real-time incident alerts and city-wide monitoring.
- Integrated 2FA security, JWT authentication, and role-based access control for admin and operator roles.
- Deployed full-stack app (React on Vercel, Node/Flask on Railway via Docker, MongoDB Atlas) , production live.
- Wrote Selenium test suite for QA coverage and conducted full STLC cycle including bug reporting in Jira AIO.

FoodieExpress – Full-Stack Food Delivery App | MongoDB, Express, React, Node.js (MERN), JWT

- Built full MERN stack app with JWT auth, protected routes, restaurant browsing, cart persistence, and admin dashboard.
- Implemented role-based access for customers and restaurant admins with secure session management.

MindMate – AI Learning & Feedback Platform | React, Node.js, Hugging Face APIs, MVC Architecture

- Built AI-powered learning platform with automated quiz generation and document feedback via Hugging Face APIs.
- Designed clean UI with MVC architecture and RESTful API integration for dynamic content delivery.

SADSS – Agriculture Decision Support System | Python, Decision Tree, KNN, Linear Regression, Tkinter

- Developed ML-based crop recommendation system using three classification algorithms with a Tkinter GUI.
- Implemented data preprocessing, model training, and comparative accuracy analysis across algorithms.

Wonderland ToyStore – E-Commerce Backend | Spring Boot, MySQL, Stripe API, MVC

- Developed secure REST API backend with Stripe payment integration, inventory management, and full cart operations.

Firooz / Lumière – E-Commerce Frontend | React, Vite, Tailwind CSS, Framer Motion

- Built polished e-commerce UI with product catalog, cart/checkout flow, and smooth Framer Motion animations.
- Applied responsive design principles and HCI best practices for intuitive user experience.

ZipGo – Ride-Sharing Desktop App | JavaFX, MySQL, Scene Builder

- Designed JavaFX desktop application with booking management, fare calculation, real-time chat, and user feedback.

SOFTWARE QUALITY ENGINEERING WORK

- Designed and executed Selenium WebDriver test suites for SafeCity AI and FYP Management System web applications.
- Managed test cases, defect tracking, and test execution reports using Jira AIO plugin.
- Wrote complete STLC documentation including test plans, test cases, and bug reports.

LANGUAGES

English (Professional) | **Urdu** (Native)